

CROSSSAFE · REQUIREMENTS

CrossSafe requirements

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DOCUMENT CONTROL

Document ID	CRS-REQ-001	Version	0.5
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/crosssafe		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-30	Amish Chadha	Status from CRS-CAL-001 v0.3 for the constructable design (CRS-DDR-003); R15 now not met on paper
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from CRS-CAL-001; R15 redefined to the existing-pole assembly (CRS-DDR-001 D1); R1, R9 and R11 updated for D3, D4 and D8
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

CrossSafe requirements

These requirements are checked by calculation in CRS-CAL-001 at TRL 3. Targets are still proposals, not yet validated with users or a road authority, and will be revised after co-design (see CRS-PRB-001). Where a local rule is stricter, the local rule wins. The design choices behind them were accepted by Amish on 2026-09-25 (CRS-DDR-001 and CRS-DDR-002).

The **design case** is a marked crossing on a two-lane road 7 m kerb to kerb, one CrossSafe assembly on each side, 300 activations of 20 s per day, 2.5 peak sun hours per day in the worst month, and an ambient range of -20 to +50 °C.

Table 1. Requirements and status at TRL 3 (CRS-CAL-001 v0.3, Table 5).

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
R1	Warning light geometry and pattern	Two amber rectangular indications per face, each at least 127 x 51 mm (5 x 2 in); the FHWA IA-21 wig-wag sequence at 75 flashing sequences per minute; pattern set in firmware for other jurisdictions; an optional pedestrian pilot light where allowed	Design review against the pilot jurisdiction's rules	Met by design (140 x 62 mm lenses)
R2	Visibility	Indications distinguishable by a driver at 100 m or more in direct sun (IA-21 cites SAE J595 Class 1 yellow peak intensity) and not glaring at night (automatic dimming)	Photometric calculation from LED data; later field check	Not verifiable at TRL 3 (screening: about 3,939 cd per head)
R3	Activation	Push button and passive presence detection; both sides of the road flash within 0.5 s of a trigger on either side	Timing budget; later bench test	Met on paper (48 ms; 244 ms with two retries)
R4	Flash duration	Set per site from crossing length / 0.9 m/s + 5 s, adjustable 10 to 40 s; a new trigger while flashing extends the time	Firmware review	Met by design (12.8 s for 7 m)
R5	Detection quality	Passive detection triggers for a person waiting 1 s or more in the waiting zone; false activations 5 % or fewer of all activations; missed waiting pedestrians 2 % or fewer	Site trial with manual counts	Not verifiable at TRL 3
R6	Autonomy without sun	5 days or more of design-case use from a full battery	Energy calculation	Met on paper (7.7 days; 5.4 at -20 °C)
R7	Energy balance	Energy neutral at 2.5 peak sun hours; recover from 20 % to full in 3 days or fewer at 5 peak sun hours	Solar yield calculation	Met on paper (break-even 1.05 h; recovery 2.1 days)
R8	Environment	Enclosures and connectors IP65 or better; operate -20 to +50 °C ambient; battery charges only at 0 to 45 °C cell temperature	Datasheets and heat balance; later spray and thermal tests	Met on paper with the sun shield (53.9 °C inside at 50 °C ambient when dusty; charging to 41.1 °C ambient)
R9	Structure	Post and clamps carry the sign and panel in a 40 m/s gust with stress 60 % or less of yield; new posts are 114.3 x 3.6 mm (CRS-DDR-001 D3); the sign cannot rotate on its pole (anti-rotation bolt on new posts, keyed saddles on existing poles, CRS-DDR-002)	Wind load calculation	At risk on poles under about 70 mm only: post 45 % of yield and bolt 12 % of bearing (met); keyed saddles resist 151 N·m on a 60 mm pole against 175 N·m
R10	Mounting	Fits the project's own post or an existing 60 to 114 mm pole with band clamps; sign bottom at 2.1 m or more (or local rule); two-person crew installs one assembly in 2 h or less	Design review; later timed trial	Not verifiable at TRL 3 (fit met; install estimate 120 min, at the limit)
R11	Accessibility	Push button 0.9 to 1.1 m above the walking surface, operable with 22 N or less, with an	Design review with disability groups	Met by design (button at 1.05 m), to be checked in co-design

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
		audible and tactile acknowledgement and a pedestrian-facing pilot light		
R12	Fail-safe behavior	Never flashes continuously; a fault (low battery, open LED, lost radio link) is shown on a maintenance indicator and logged; with the link lost, each side still flashes on a local trigger	Firmware review; fault injection later	Met by design
R13	Privacy	No images or audio are recorded or leave the device; only activation counts, faults and battery state are logged or sent	Design review	Met by design
R14	Theft and vandal resistance	Battery, charger and controller 2.8 m or more above the walking surface; security fasteners; no exposed cables below 2.5 m	Design review	Met by design (enclosure bottom 2.85 m; sensing head 2.515 m and up)
R15	Affordable	Parts for one assembly on an existing pole \$350 or less (the prototype <code>budget_usd</code> covers, CRS-DDR-001 D1); parts for a two-sided crossing on existing poles \$700 or less. Crossings that need new posts are costed and reported, not held to this target	Priced BOM (<code>bom/bom.csv</code>)	Not met on paper (\$360.00 per assembly, \$10.00 over; \$720.00 per crossing, \$20.00 over); new posts \$433.00 and \$866.00

■ Requirements at risk or not verifiable

One requirement is not met on paper (R15, cost), one is at risk and three need evidence that a paper study cannot give.

- **R15 not met on paper after the design for construction (CRS-DDR-003).** The saddles, mounting plate, panel bracket, radar arm, battery strap and fixings that make every part buildable and fixed add \$28.00: one assembly on an existing pole is \$360.00 against \$350, and a crossing \$720.00 against \$700. Raising `budget_usd` to \$375 is proposed, awaiting Amish (design decisions register, item 1).
- **R9 at risk on small poles.** The 114.3 x 3.6 mm post meets the stress target at 45 % of yield, and a 1,800 mm embedment in a 500 mm footing is a screening size. On new posts the M10 through-bolt stops the sign rotating with a wide margin. On existing poles the keyed saddles hold on poles of about 70 mm and larger at an assumed grip friction of 0.4; below that the sign could still slip. A third band on small poles is proposed, awaiting Amish (CRS-DDR-002, N1). On existing poles the host pole must carry about 2.5 kN·m more at the sidewalk, which the owner must confirm.
- **R8 met on paper after the sun shield (CRS-DDR-002).** Unshielded, a dusty enclosure ran 14.7 K above ambient and reached 64.7 °C at 50 °C ambient; with the ventilated shield the rise is 3.9 K and the interior 53.9 °C, below the typical 60 °C discharge limit. The shield factor is borrowed, not measured.
- **R2, R5 and R10 are not verifiable at TRL 3.** R2 and R5 decide whether drivers respond and whether passive detection is trusted.
- **R6, not met at TRL 2, is now met on paper after the PIR gating (D2).**

■ Assumptions

The full list is in CRS-CAL-001, Table 1. The main ones are:

- LED heads draw about 6 W each when lit in daytime and are lit 25 % of each IA-21 sequence; LED drivers are about 91 % efficient.
- Standby about 187 mW: radio in continuous receive and controller about 23 mW, 12 V quiescent about 13 mW, charger about 51 mW, radar 0.4 W on 25 % of the day under PIR gating.
- Solar system efficiency about 76 % from panel rating to energy stored (panel heat and dust 0.85, MPPT 0.94, cell charge 0.95).
- Walking speed of 0.9 m/s allows for children and older people; the US MUTCD basis is faster, and local rules decide.
- Gust wind speed of 40 m/s and a drag coefficient of 1.2 for every part; S235 steel for the post; band tension 1,000 N; keyed saddle grip friction 0.4; a ventilated shield passes 25 % of the solar gain.